

DTU



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- It is only for widening your horizons on the topic.

Lazaros Nalpantidis

Stereo Vision

~~Cherry on Top~~



Outline

- What is Stereo Vision?
- Stereo/Epipolar Geometry
- Rectified Stereo Case
- Depth from Stereo Matches
- Correspondence Problem
 - **Dense** vs Sparse Correspondence
 - **Local** vs Global Correspondence
 - (Dis)-Similarity Measures
- Summary

One more thing...



Lightweight Monocular Depth Estimation through Guided Decoding

- M. Rudolph, Y. Dawoud, R. Güldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," *IEEE ICRA*, 2022.

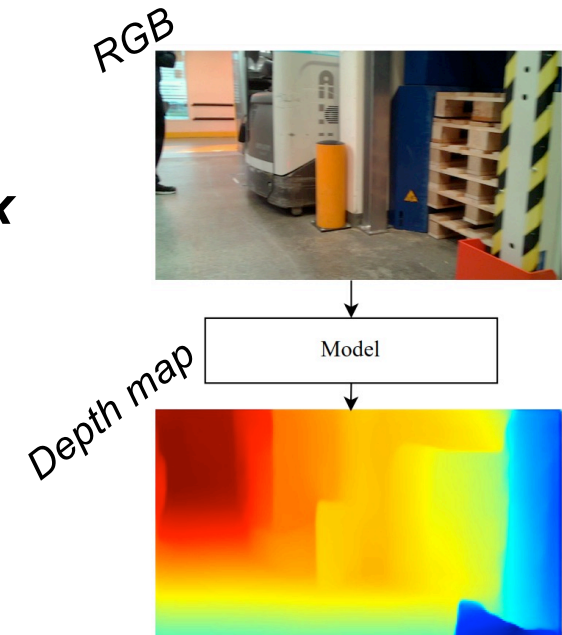
Lightweight Monocular Depth Estimation through Guided Decoding

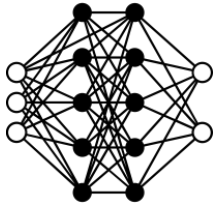
Monocular Depth Estimation (MDE)

- Predict a **depth map y** from a **single input image x**
- **Inexpensive** compared to active sensors
- Recent advances thanks to **Deep Learning**

However:

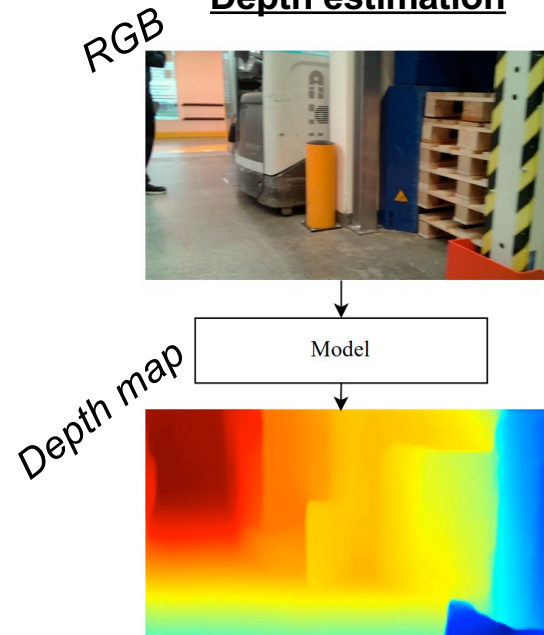
- Large models with **many parameters**
→ Slow Inference
- We need **lightweight** and **accurate** models for robotic applications





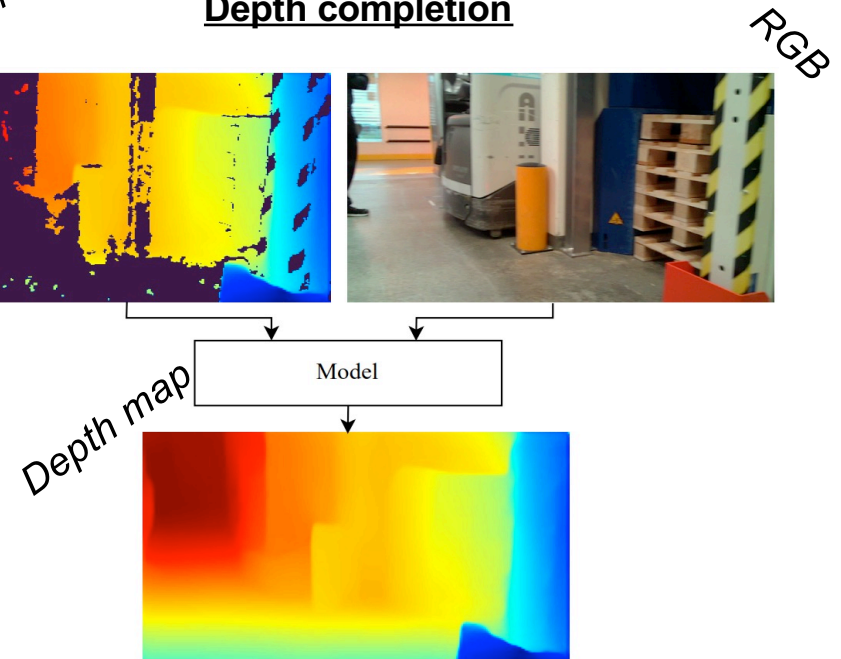
Use deep learning methods to infer depth of the scene and move from 2D to 3D.

Depth estimation



Sparse depth map

Depth completion



Lightweight Monocular Depth Estimation through Guided Decoding

Model operating on higher resolutions

- Capture details in the depth maps
- Balance accuracy and speed on NVIDIA Jetson devices

Easy deployment

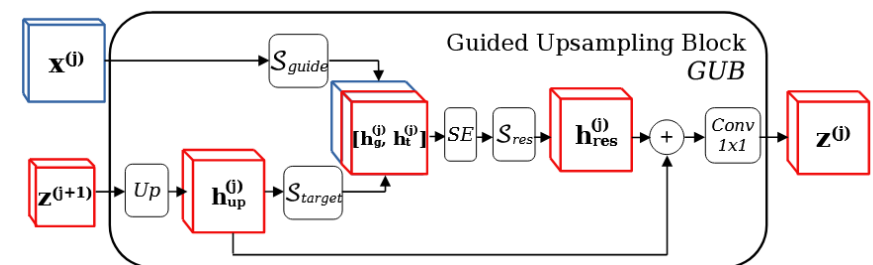
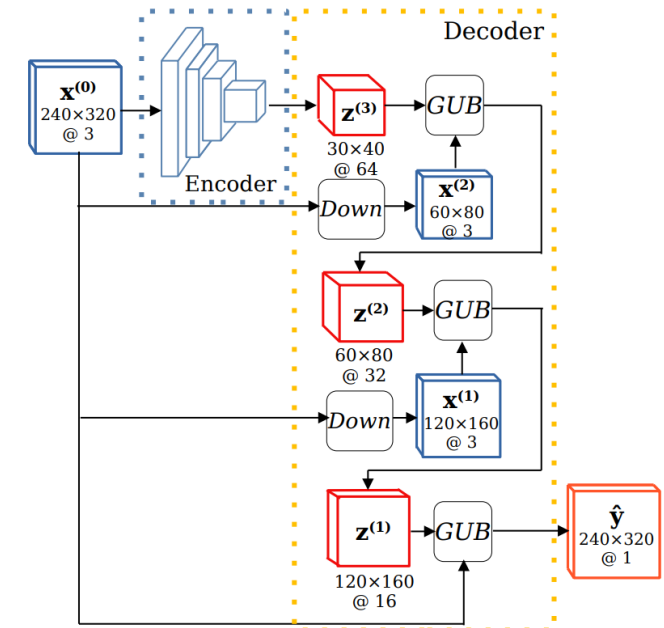
- Use tensorRT for inference

Use DDRNet as encoder

- Designed for real-time Semantic Segmentation

Design a Decoder based on feature guidance

- Use input image to guide the up-sampling process in the Decoder
- Allows to recover fine-grained details in the feature maps



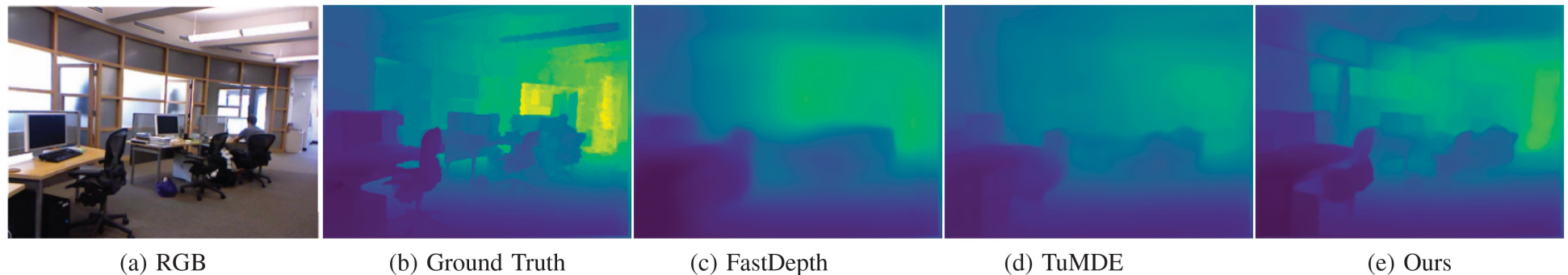
- M. Rudolph, Y. Dawoud, R. Gldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," *IEEE ICRA*, 2022.

Lightweight Monocular Depth Estimation through Guided Decoding

- KITTI dataset:



- NYU v2 dataset:



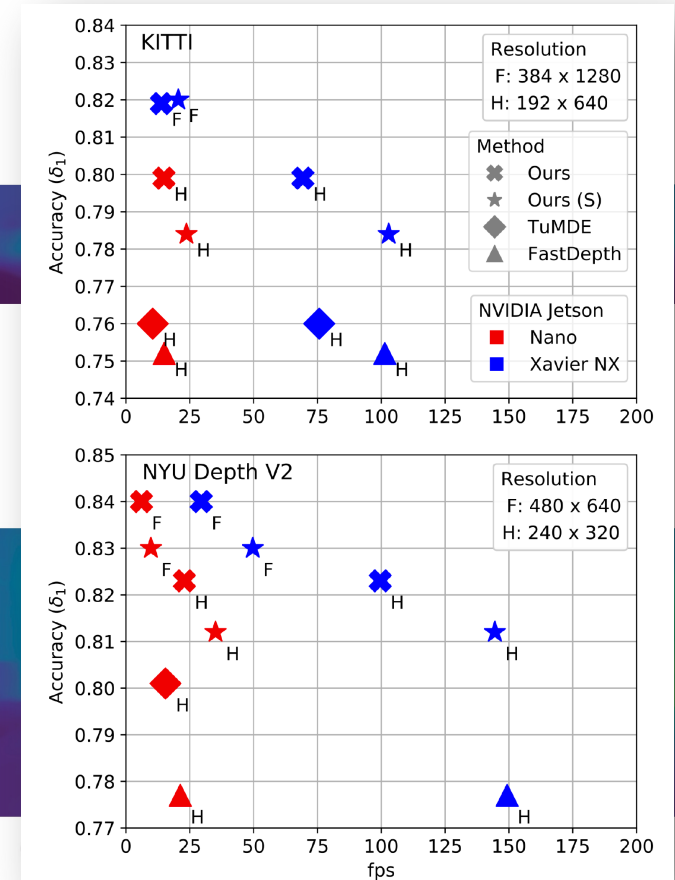
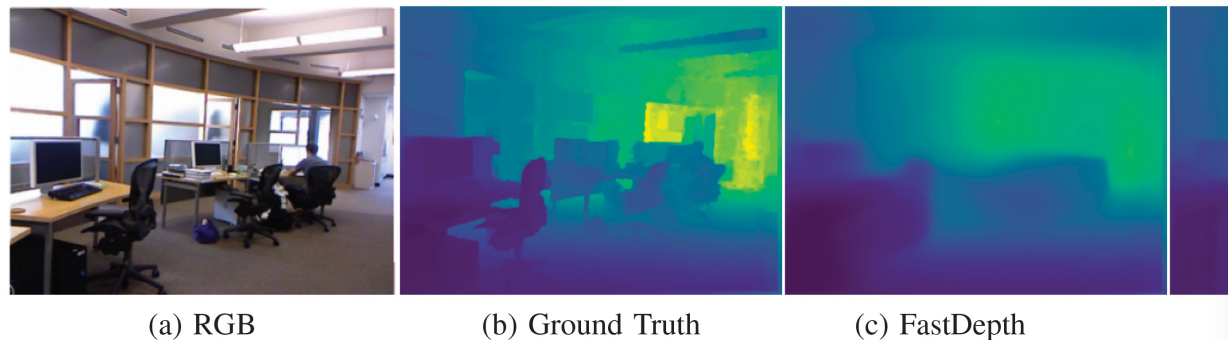
- M. Rudolph, Y. Dawoud, R. G ldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," *IEEE ICRA*, 2022.

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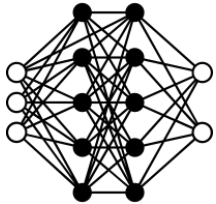
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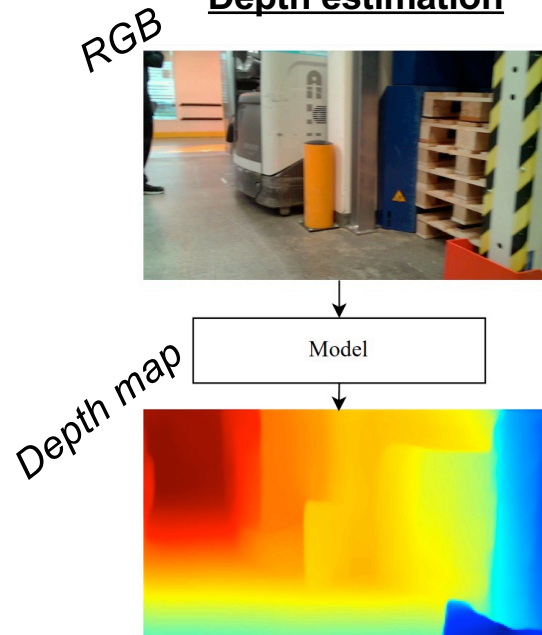


- M. Rudolph, Y. Dawoud, R. G ldenring, **L. Nalpantidis**, and V. Belagiannis, "Lightweight monocular depth estimation through guided decoding," IEEE ICRA, 2022.



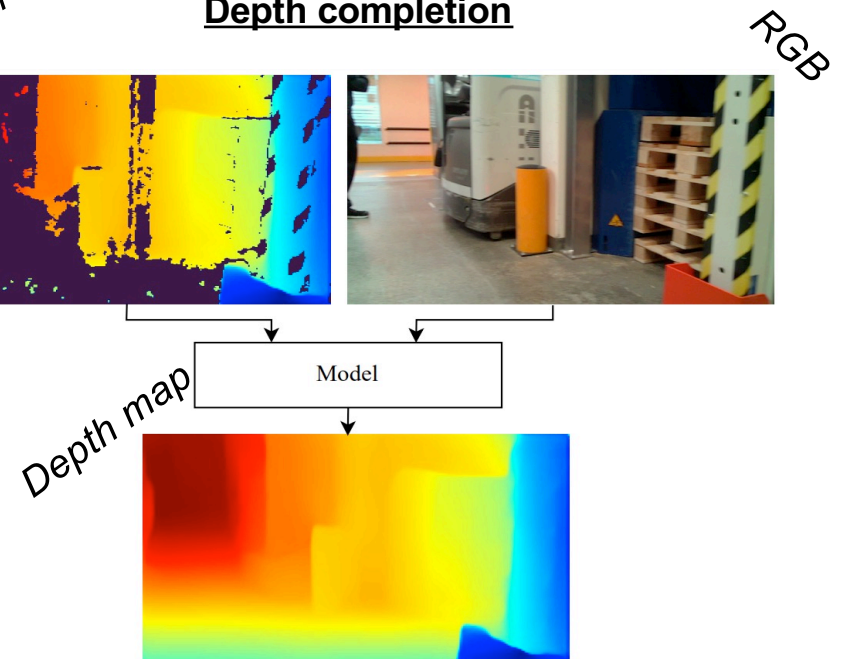
Use deep learning methods to infer depth of the scene and move from 2D to 3D.

Depth estimation



Sparse depth map

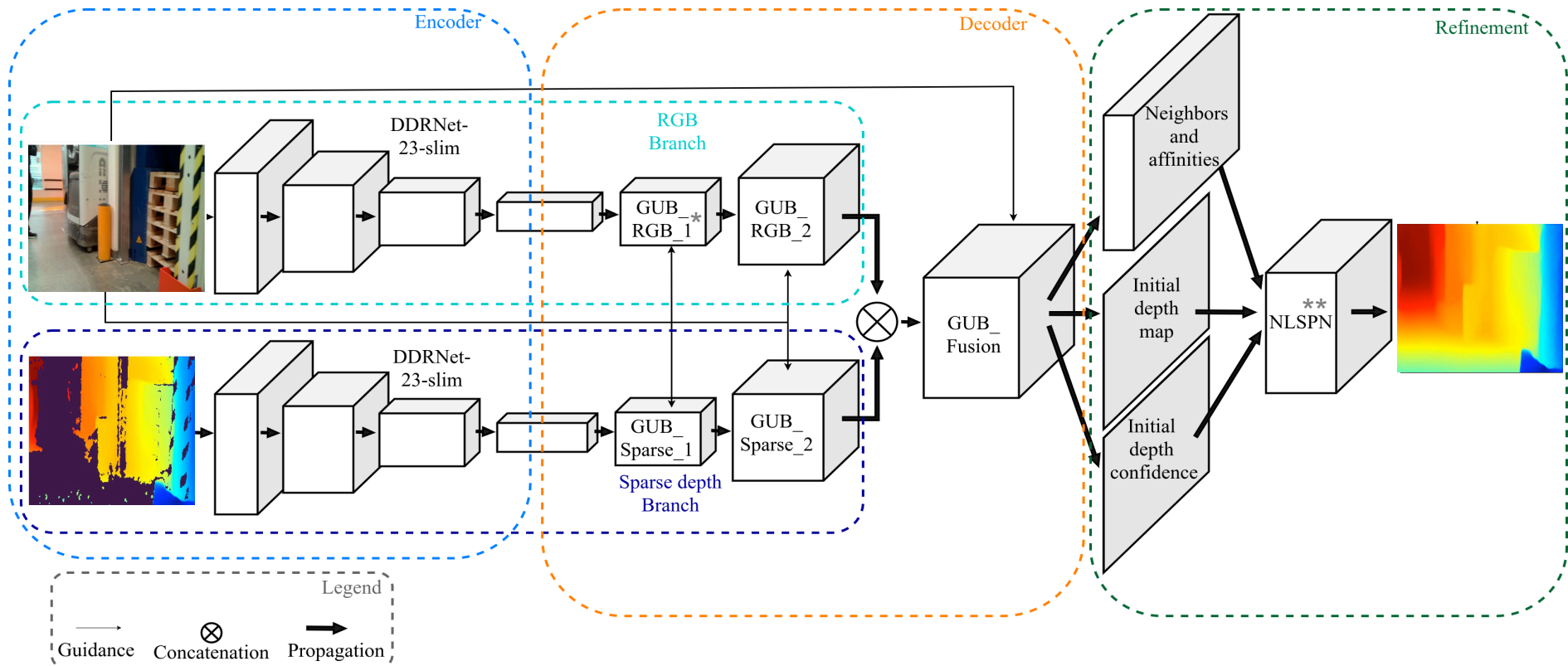
Depth completion





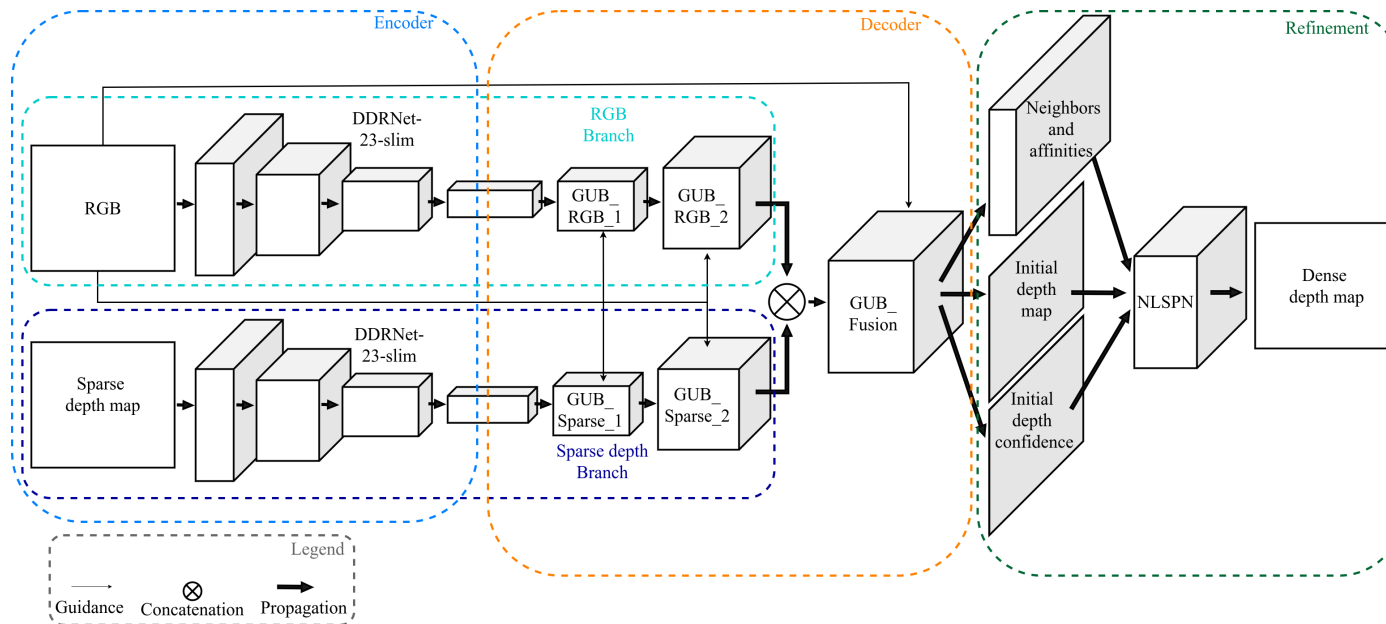
Goals:

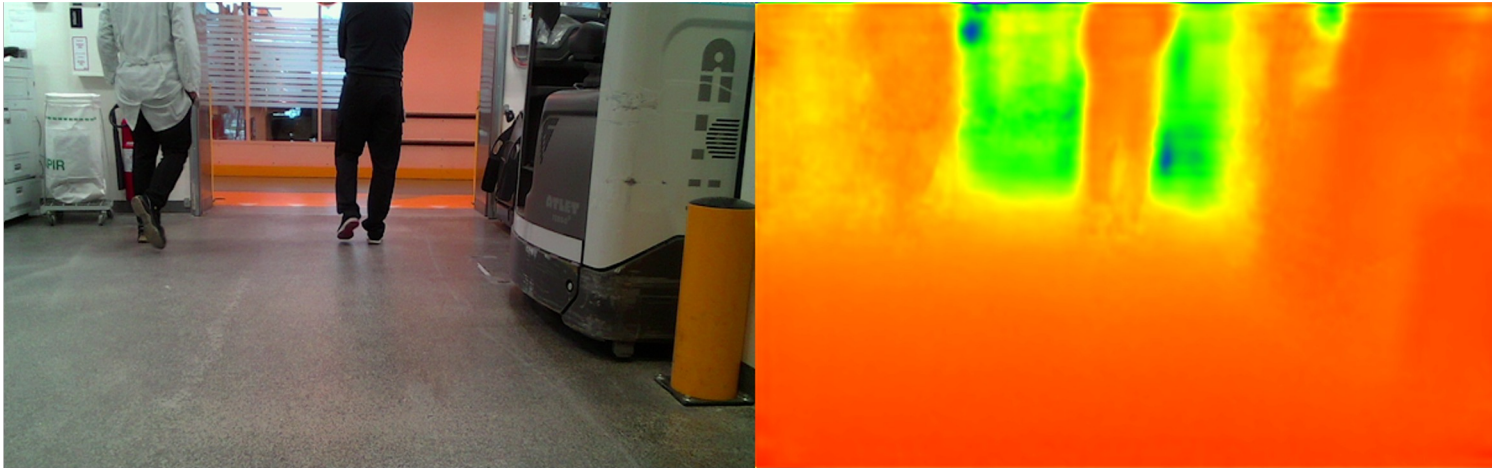
- Depth estimation model that can benefit from any available depth data → depth completion
- The model must be suitable for resource-constrained devices



- D. Arapis, M. Jami, and **L. Nalpantidis**, "Bridging depth estimation and completion for mobile robots reliable 3D perception," in *International Conference on Robot Intelligence Technology and Applications*, (Gold Coast, Australia), 2022.

Method	RMSE↓	REL↓	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	MACs(G)↓
(Our) LateFusion+NLSPN	0.099	0.014	99.4	99.9	99.9	10.7
LateFusion	0.116	0.018	99.2	99.8	99.9	7.1
EarlyFusion	0.114	0.018	99.3	99.8	99.9	10.0
EarlyFusion+NLSPN	<u>0.107</u>	<u>0.015</u>	<u>99.3</u>	<u>99.8</u>	<u>99.9</u>	13.6
Small	0.165	0.033	98.4	99.7	99.9	3.1
(Our_small) Small+NLSPN	0.111	0.017	99.3	99.8	99.9	4.0





Key findings



It is possible to **bridge depth estimation and completion** by designing networks that can incorporate **any amount of depth values to improve** their prediction.



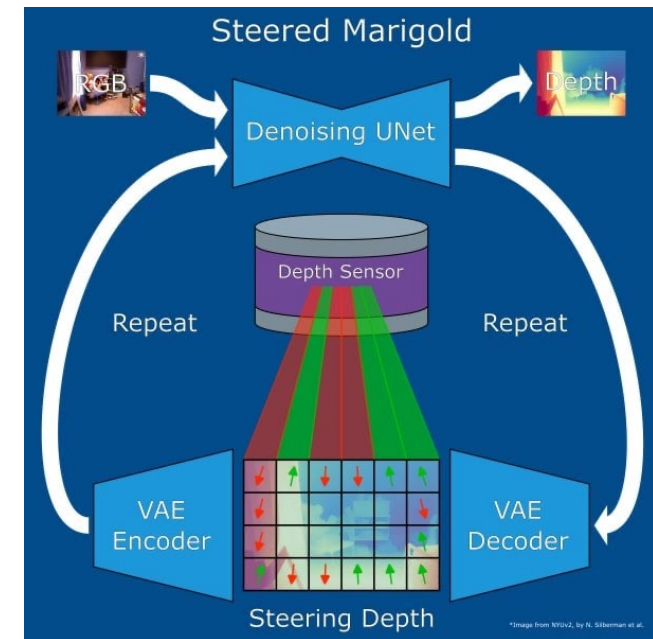
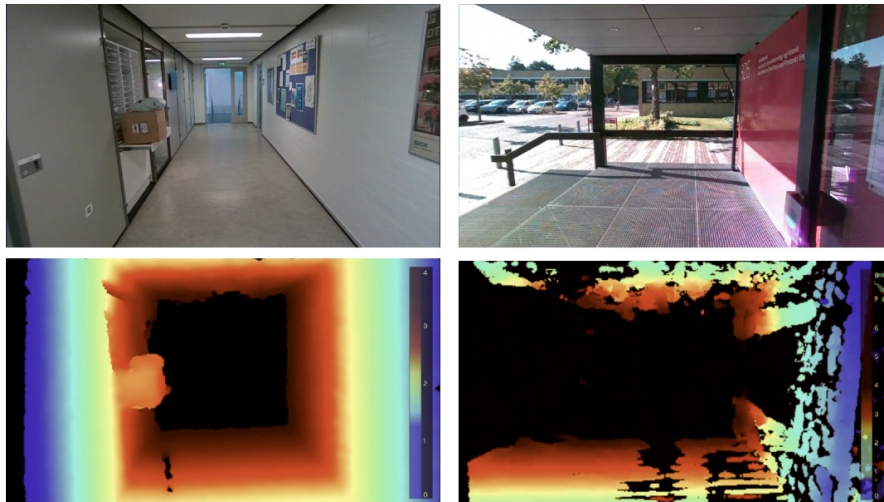
The proposed **refinement method** has a **bigger impact** in improving predictions compared to more complex backbones.



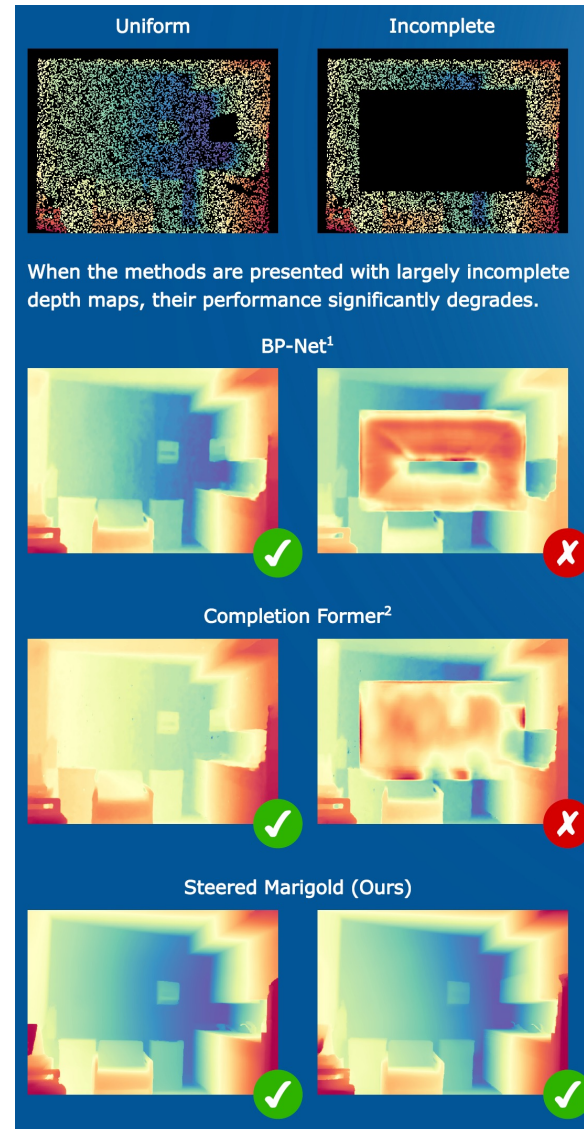
SteeredMarigold: Steering Diffusion Towards Depth Completion of Largely Incomplete Depth Maps

- *Jakub Gregorek and Lazaros Nalpantidis, "SteeredMarigold: Steering Diffusion Towards Depth Completion of Largely Incomplete Depth Maps," IEEE ICRA, 2025.*

SteeredMarigold



SteeredMarigold



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